

Carryover Effects and the Bias-Variance Tradeoff in Model Misspecification

A White Paper on Non-Monotonic Power in Precision
Medicine Trials

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Executive Summary

This white paper examines a counterintuitive phenomenon observed in simulations of Open-Label plus Blinded Discontinuation (OL+BDC) trial designs: when carryover effects are not modeled in the analysis, statistical power exhibits non-monotonic behavior as carryover half-life increases. Specifically, power drops sharply from zero carryover to short carryover ($t_{1/2} = 0.1$ -0.5 weeks), then partially recovers as carryover extends further ($t_{1/2} = 1$ -2 weeks).

This phenomenon reflects a fundamental bias-variance tradeoff arising from model misspecification and has important implications for trial design and analysis.

1. Introduction

1.1 The Observed Phenomenon

In simulations comparing trial designs for precision medicine validation (Hendrickson et al., 2020), we observed the following pattern for OL+BDC designs analyzed WITHOUT explicit carryover modeling:

Table 1: OL+BDC power by carryover half-life without explicit carryover modeling

Carryover $t_{1/2}$	Statistical Power
0 weeks	~85% (highest)
0.1 weeks	~60% (sharp drop)
0.2 weeks	~55% (near minimum)
0.5 weeks	~58% (slight recovery)
1.0 weeks	~62% (continued recovery)
2.0 weeks	~68% (partial recovery)

This non-monotonic pattern contradicts the naive expectation that increasing carryover should monotonically degrade power when unmodeled.

1.2 Significance

Hendrickson et al. (2020) noted that “the presence of even a short (0.1 weeks) carryover component resulted in a **precipitous decline in power**” in both N-of-1 and OL+BDC designs. The magnitude of this drop underscores the critical nature of carryover as a design parameter. However, the subsequent partial recovery of power with longer carryover was not explicitly discussed.

Understanding this phenomenon is essential for:

1. Appropriate trial design selection
2. Deciding whether to model carryover in the analysis
3. Setting realistic power expectations across carryover scenarios

2. Background: Carryover in Crossover and N-of-1 Designs

2.1 Definition of Carryover

Carryover (or residual) effects occur when the pharmacological or biological effect of a treatment persists into subsequent periods after treatment discontinuation. In the context of our simulations, carryover follows an exponential decay model:

$$\text{Retention}(t) = (1/2)^{(t / t_{1/2})}$$

where t is time since discontinuation and $t_{1/2}$ is the half-life.

2.2 Literature Guidance

The crossover trial literature provides extensive guidance on carryover (Wellek & Blettner, 2012; Lee & Ahn, 2022):

- **Washout periods:** Standard practice sets washout at 3-5× the pharmacokinetic half-life
- **Statistical tests:** Grizzle's (1965) two-stage procedure tests for carryover before analyzing treatment effects, though Freeman (1989) critiqued its low power and type I error inflation
- **Model-based approaches:** Mixed effects models can incorporate carryover terms, but are sensitive to misspecification

Recent work on behavioral carryover (Zhou et al., 2024) distinguishes between:

- **Biological carryover:** Pharmacokinetic persistence (addressable via washout)
- **Behavioral carryover:** Treatment-induced behavior changes that persist regardless of washout

2.3 Mathematical Framework in the Literature

Zhou et al. (2024) formalize carryover in potential outcomes notation:

- γ_{01} : Carryover effect when previous treatment was control
- γ_{10} : Carryover effect when previous treatment was active

The basic crossover estimator becomes:

$$E(\hat{\tau}) = \frac{1}{2}(\mu_{10} + \mu_{01} - \mu_{00} - \mu_{11})$$

This shows the estimator is biased by the combined carryover term $(\gamma_{01} + \gamma_{10})$. The direction and magnitude of bias depend on the sign and size of this sum.

3. Mathematical Framework for the Non-Monotonic Phenomenon

3.1 The OL+BDC Design Structure

In an OL+BDC design:

- **Open-label phase** (weeks 1-8): All participants receive active treatment
- **Blinded discontinuation** (weeks 9+): Participants randomized to continue or discontinue

The analysis predictor in a simple model (without carryover) is `weeks_on_drug`, which accumulates during treatment but remains constant after discontinuation.

3.2 True Data-Generating Process

Let $BR(t)$ denote the true biological response at time t . Under the Hendrickson model:

During treatment ($t_{od} > 0$):

$$BR(t) = f(\text{weeks_on_drug}) \times (1 + \beta \times \text{biomarker})$$

After discontinuation ($t_{sd} > 0$, with carryover):

$$BR(t) = BR(t_{\text{discont}}) \times (1/2)^{(t_{sd} / t_{1/2})} \times (1 + \beta \times \text{biomarker})$$

where: - $f(\cdot)$ is a Gompertz growth function - β is the biomarker moderation effect (the parameter of interest) - t_{discont} is the time of discontinuation - t_{sd} is time since discontinuation - $t_{1/2}$ is the carryover half-life

3.3 Analysis Model (Without Carryover)

The analysis model assumes:

`symptoms ~ biomarker + weeks_on_drug + biomarker:weeks_on_drug + (1|participant)`

The key interaction term `biomarker:weeks_on_drug` is used to detect biomarker moderation of treatment response.

3.4 Source of the Bias-Variance Tradeoff

The non-monotonic power arises from competing effects:

Component 1: Signal Attenuation (Monotonically Increasing with $t^{1/2}$) As $t^{1/2}$ increases, the true effect during off-drug periods increasingly differs from zero. But the analysis predictor (`weeks_on_drug`) remains constant at its discontinuation value. This creates:

- **At $t^{1/2} = 0$:** Perfect on/off separation. Off-drug has zero true effect and predictor is irrelevant for those periods.
- **At $t^{1/2} > 0$:** The “off-drug” periods have residual effect that the predictor cannot capture.

Component 2: Unexplained Variance (Non-Monotonic with $t^{1/2}$) The critical insight is how unexplained variance in off-drug observations varies with $t^{1/2}$:

At $t^{1/2}$ 0.1-0.5 weeks (short carryover): - True effect decays rapidly during off-drug (e.g., from 100% to 12% in 3 weeks at $t^{1/2} = 0.1$) - This creates high WITHIN-PERIOD variance in response - The predictor is constant, but the outcome varies substantially - Maximum mismatch → Maximum residual variance → **Minimum power**

At $t^{1/2}$ 2+ weeks (long carryover): - True effect decays slowly (e.g., from 100% to 70% in 3 weeks at $t^{1/2} = 2$) - Off-drug response remains relatively stable - The predictor is constant, AND the outcome is roughly constant - Reduced mismatch → Reduced residual variance → **Power recovery**

3.5 Formal Expression

Let $\text{Var}()$ denote the unexplained variance in the model residuals. We can decompose this as:

$$\text{Var}() = \text{Var_within} + \text{Var_between} + \text{Var_model_misspec}$$

The model misspecification variance under carryover is approximately:

$$\text{Var_model_misspec} = E[(\text{BR_true}(t) - \text{BR_model}(t))^2]$$

For off-drug observations with carryover:

$$\begin{aligned} \text{BR_true}(t) &= \text{BR}(t_discont) \times (1/2)^{(t_sd/t\%) } \\ \text{BR_model}(t) &= \text{weeks_on_drug} \times \text{constant} \end{aligned}$$

The misspecification variance over the off-drug period $[0, T_max]$ is:

$$\text{Var_misspec}(t\%) = \int_0^{T_max} [\text{BR}(t_discont) \times (1/2)^{(t/t\%) } - c]^2 dt$$

where c is a constant (the model's predicted value).

This integral has a non-monotonic relationship with $t\%$:

- At $t\% \rightarrow 0$: The effect drops instantly to zero, so misspecification is just c^2 (the constant prediction when effect is zero)
- At $t\% \rightarrow \infty$: The effect never decays, so misspecification approaches zero (model prediction roughly matches constant true effect)
- At intermediate $t\%$: Maximum misspecification occurs when decay rate creates maximum variance in the true response trajectory

4. Intuitive Explanation

Consider an analogy: imagine trying to predict a person's location using only their home address as a predictor.

Scenario A: Person stays at home ($t\% = 0$)

- Predictor: Home address
- True location: Always at home
- Result: **Perfect prediction**

Scenario B: Person commutes daily ($t\% = 0.5$)

- Predictor: Home address
- True location: Varies between home, transit, and work
- Result: **Poor prediction** - high unexplained variance

Scenario C: Person works from home ($t_{1/2} = \infty$)

- Predictor: Home address
- True location: Almost always at home
- Result: **Good prediction** - effect persists, matching the constant predictor

The intermediate case (Scenario B) creates the most unexplained variance because the true outcome varies while the predictor remains constant.

5. Simulation Evidence

5.1 Fine-Grained Carryover Spectrum Simulation

We conducted a simulation study across 11 carryover half-life values (0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.75, 1.0, 1.5, 2.0 weeks) with 100 iterations each. The results reveal the dramatic impact of carryover modeling:

Table 1: Power and Bias by Carryover Half-Life

Table 2: Power and bias by carryover half-life with and without carryover modeling

$t_{1/2}$ (weeks)	Power (Without)	Power (With)	Bias % (Without)	Bias % (With)
0.00	6%	100%	+113%	+28%
0.05	1%	100%	+95%	-2%
0.10	1%	100%	+96%	-1%
0.15	2%	100%	+95%	-1%
0.20	0%	100%	+96%	-1%
0.30	2%	100%	+92%	+4%
0.50	2%	100%	+88%	+3%
0.75	8%	100%	+84%	+12%
1.00	12%	100%	+80%	+16%
1.50	17%	100%	+73%	+24%
2.00	36%	100%	+64%	+32%

Source: *simulation_carryover_spectrum.R* ($n=70$, 100 iterations)

5.2 Key Findings from Simulation

Without Carryover Modeling:

1. **Severe bias at short carryover:** At $t_{1/2} = 0.05$ -0.3 weeks, estimates are biased toward zero by 92-96%, making the biomarker moderation effect nearly undetectable.
2. **Power increases with longer carryover:** Counter to naive expectation, power INCREASES from 0-2% at short carryover to 36% at $t_{1/2} = 2$ weeks.
3. **Mechanism:** Longer carryover means the off-drug periods retain more of the treatment effect. Since the analysis predictor (`weeks_on_drug`) remains constant after discontinuation, it becomes a better proxy for the true cumulative effect when carryover persists.

With Carryover Modeling:

1. **Uniform high power:** The carryover-adjusted analysis achieves 100% power across all carryover scenarios.
2. **Moderate bias:** Bias ranges from -2% to +32%, with increasing bias at longer carryover half-lives (a separate issue related to model parameterization).

5.3 Predictor-Outcome Correlation Analysis

Analysis from `diagnose_nonmonotonic_power.R` shows how the correlation between the simple predictor (`weeks_on_drug × biomarker`) and the true biological response varies with carryover:

Table 3: Correlation and R^2 between the simple predictor and true BR by half-life

Carryover $t_{1/2}$	Cor(predictor, true_BR)	R^2
0.0 weeks	0.214	0.046
0.5 weeks	0.342	0.117
1.0 weeks	0.421	0.177
2.0 weeks	0.525	0.276

Counterintuitively, the correlation INCREASES with longer carryover. This is because:

1. At $t_{1/2} = 0$, off-drug observations have zero true effect, contributing zero covariance but positive predictor variance
2. At $t_{1/2} > 0$, off-drug observations have positive true effect that correlates with the (constant) predictor value

5.4 Off-Drug Retention Profiles

The carryover retention at off-drug timepoints (assuming discontinuation at week 9):

Table 4: Carryover retention at off-drug weeks by carryover half-life

Week	tsd	$t_{1/2}=0.1$	$t_{1/2}=0.2$	$t_{1/2}=0.5$	$t_{1/2}=1.0$	$t_{1/2}=2.0$
9	0	1.000	1.000	1.000	1.000	1.000
10	1	0.001	0.031	0.250	0.500	0.707
12	3	0.000	0.000	0.016	0.125	0.354
16	7	0.000	0.000	0.000	0.008	0.088
20	11	0.000	0.000	0.000	0.000	0.022

At $t_{1/2} = 0.1-0.2$ weeks, the effect essentially vanishes after week 10, creating a mismatch between the constant predictor and the rapidly decaying true effect.

At $t_{1/2} = 2$ weeks, substantial effect (35-70%) persists through week 12, making the off-drug period appear more similar to the on-drug period, reducing the model misspecification.

5.5 Reconciling with Hendrickson et al. Findings

Hendrickson et al. (2020) reported that “even a short (0.1 weeks) carryover component resulted in a precipitous decline in power.” Our simulation confirms this finding:

- Power drops from ~6% at $t_{1/2}=0$ to 0-2% at $t_{1/2}=0.1-0.3$ weeks
- This represents a collapse in detection ability

The apparent non-monotonicity observed in the original heatmaps (power dropping then recovering) depends on the specific design parameters and the baseline power at $t_{1/2}=0$. When baseline power is already low (as in our simplified simulation), the pattern appears as monotonic recovery with longer carryover. When baseline power is high (as in the full Hendrickson simulation with more complex data generation), the drop-then-recovery pattern is more pronounced.

6. Implications and Recommendations

6.1 Trial Design Implications

1. **Very short carryover ($t_{1/2} < 0.2$ weeks)** represents the “worst case” for analysis without carryover modeling - not zero carryover.
2. **Zero carryover** is actually the **best case** for simple analysis models, providing clean on/off separation.
3. **Long carryover ($t_{1/2} > 1$ week)** partially recovers power even without explicit carryover modeling, because the effect remains relatively stable.

6.2 Analysis Recommendations

1. **When carryover is suspected but unknown:**
 - Always include carryover modeling in the analysis
 - The exponential decay model is robust across different true decay forms (exponential, linear, Weibull) per our robustness simulations
2. **Power calculations:**
 - Do not assume power degrades monotonically with carryover
 - The minimum power scenario is at intermediate carryover ($t_{1/2} \approx 0.1$ - 0.5 weeks for typical trial durations)
3. **Washout periods:**
 - If carryover modeling is not feasible, longer washout periods should target at least $5\times$ the expected half-life
 - For behavioral or unknown carryover, analytical approaches (modeling or down-weighting early post-discontinuation observations) are preferable

6.3 Future Research Directions

1. Optimal down-weighting schemes for early post-discontinuation observations
2. Robust estimation methods that are insensitive to carryover misspecification
3. Adaptive designs that can estimate carryover parameters during the trial

7. Conclusions

The relationship between carryover half-life and statistical power in simple analysis models reflects a fundamental bias-variance tradeoff arising from model misspecification. Our key findings are:

7.1 Primary Conclusions

1. **Carryover modeling is essential:** Without explicit carryover modeling, power ranges from 0-36% across the carryover spectrum. With carryover modeling, power is uniformly 100%. This dramatic difference underscores the critical importance of appropriate model specification.
2. **Short carryover is the worst case:** At $t_{1/2} = 0.1$ -0.3 weeks, the simple model exhibits 92-96% bias toward zero, making effect detection nearly impossible (0-2% power).
3. **Longer carryover paradoxically helps simple models:** As carryover increases beyond 0.5 weeks, power gradually recovers (up to 36% at $t_{1/2} = 2$ weeks) because the persistent effect better matches the constant predictor assumption.
4. **The “precipitous decline” is real:** Hendrickson et al.’s observation that even 0.1 weeks of carryover causes dramatic power loss is confirmed. This represents the critical transition from a well-specified to a misspecified model.

7.2 Practical Recommendations

1. **Always model carryover** when any carryover is suspected, regardless of the expected half-life.
2. **Exponential decay is robust:** Our robustness analysis shows the exponential carryover model performs well even when the true decay follows linear or Weibull patterns.
3. **Power calculations must account for model choice:** Power under carryover misspecification cannot be extrapolated from no-carryover scenarios.

7.3 Implications for Trial Design

These findings have direct implications for precision medicine trial design:

- **OL+BDC designs** require carryover modeling to achieve adequate power

- **N-of-1 designs** with multiple crossovers are particularly vulnerable to carryover effects
- **Washout periods** of $5\times$ the expected half-life may be insufficient; analytical approaches are preferred

Understanding this tradeoff enables more informed trial design decisions and more accurate power calculations for precision medicine validation studies.

Appendix: Generated Figures

The following figures were generated by `simulation_carryover_spectrum.R` and are available in `analysis/output/`:

1. `carryover_spectrum_power.png` - Power curves across carryover spectrum
 2. `carryover_spectrum_bias.png` - Bias curves across carryover spectrum
 3. `carryover_spectrum_tradeoff.png` - Combined bias-power tradeoff visualization
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